YILDIZ TEKNİK ÜNİVERSİTESİ
İNŞAAT FAKÜLTESİ
İNŞAAT MÜHENDİSLİĞİ BÖLÜMÜ

DESIGN OF A HIGHWAY

HUSAEN ABDULGHAFOUR
19049920

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Department of Transportation Engineering

HIGHWAY ENGINEERING COURSE (INS3441)
TERMWORK

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CALCULATIONS

INTRODUCTION

This is the Pproject of the Highway engineering course (INS3441), the project involves the drawing of a highway based on given informaion. The project involves the study of the zeroline and highway route, the precise axis calculation, drawing and calculation of the clothoid for the transition curve and superelevation details. The project report also includes the precise plan, precise profile and cross-section sheets.

GIVEN

A- Digital contour Map which number is 25 (Scale: 1/2.000),

B- Project criteria:

1.	Project velocity; Design velocity (V_p)	= 70 km/h
2.	Platform width (B)	= 10 m (2 x 3,5 +2 x 1,5)
3.	Maximum longitudinal grade (s)	= % 8
4.	Right of way width	= 40 m
5.	Minimum horizontal curve length (D)	= 50 m
6.	Maximum superelevation (q_{max})	= % 8
7.	Maximum widening of the horizontal curve (b)	= 2.0 m
8.	Minimum vertical curve radius	= R_{CREST} = 4000 m = R_{SAG} = 2000 m

1- CALCULATIONS RELATED TO THE ZERO LINE STUDY

$S_0 = \frac{\Delta H_{A-C}}{L_{A-C}} \times 100 = \frac{20}{595.84} \times 100 = 3.3566\ \% < 8\%$ proper!

The elevation difference between A and C, ΔH_{A-C}, is 20 meters and the distance between them, L_{A-C}, is approximated as 595.84 meters.

$l_0 = \frac{h}{S_0} \times 100 = \frac{2}{3.3566} \times 100 = 59.58\ meters$

h is the distance between 2 contour lines on the map which is equal to 2 meters.

on the map: $l_p = l_0 \times \frac{1}{2000} = 59.58 \times \frac{1}{2000} = 0.02979\ meters = 2.979\ cm$

2- ROUTE STUDY CALCULATIONS

$|AS_1|$ = 84.72 m

$|S_1S_2|$ = 254.91 m

$|S_2C|$ = 339.24 m

Horizontal curve 1:

$\Delta_1 = 42.70°$

$R_1 = 100\ m$

$t_1 = R_1 \tan(\frac{\Delta_1}{2}) = 100 \tan(\frac{42.70}{2}) = 39.09\ m$

$b_1 = R_1(\sec\left(\frac{\Delta_1}{2}\right) - 1) = 100(\sec\left(\frac{42.70}{2}\right) - 1) = 7.37\ m$

$D_1 = \frac{2\pi \times R_1 \times \Delta_1}{360} = \frac{2\pi \times 100 \times 42.70}{360} = 74.53\ m$

Horizontal curve 2:

$\Delta_2 = 121.29°$

$R_2 = 70\ m$ (special case)

$t_2 = R_2 \tan(\frac{\Delta_2}{2}) = 70 \tan(\frac{121.29}{2}) = 124.46\ m$

$b_2 = R_2(\sec\left(\frac{\Delta_2}{2}\right) - 1) = 70(\sec\left(\frac{121.29}{2}\right) - 1) = 72.79\ m$

$D_2 = \frac{2\pi \times R_2 \times \Delta_2}{360} = \frac{2\pi \times 70 \times 121.29}{360} = 148.18\ m$

Km values of special points:

$A_{km} = 0{+}000^{00}$

$\Phi_{1,km} = A_{km} + |AS_1| - t_1 = 0.00 + 84.72 - 39.09 = 0{+}045^{63}$

$B_{1,km} = \Phi_{1,km} + \frac{D_1}{2} = 45.63 + \frac{74.53}{2} = 0{+}082^{90}$

$\Gamma_{1,km} = \Phi_{1,km} + D_1 = 45.63 + 74.53 = 0{+}120^{16}$

$\Phi_{2,km} = \Gamma_{1,km} + |S_1S_2| - t_1 - t_2 = 120.16 + 254.91 - 39.09 - 124.46 = 0{+}211^{52}$

$B_{2,km} = \Phi_{2,km} + \frac{D_2}{2} = 211.52 + \frac{148.18}{2} = 0{+}285^{61}$

$\Gamma_{2,km} = \Phi_{2,km} + D_2 = 221.52 + 148.18 = 0{+}359^{70}$

$C_{km} = |S_2C| + \Gamma_{2,km} - t_2 = 339.24 + 359.70 - 124.46 = 0{+}574^{48}$

3- KM VALUES AND BLACK ELEVATIONS

Cross section no	Km	Black elevation
A	$0+000^{00}$	900.00
1	$0+025^{00}$	901.87
$Φ_1$	$0+045^{63}$	900.63
2	$0+052^{27}$	900.00
3	$0+060^{60}$	898.00
4	$0+071^{75}$	896.58
B1	$0+082^{90}$	896.05
H1	$0+100^{00}$	893.87
5	$0+110^{08}$	893.05
$Ŧ_1$	$0+120^{16}$	892.20
6	$0+137^{92}$	890.83
7	$0+153^{09}$	892.00
K_1	$0+158^{74}$	892.83
8	$0+166^{49}$	892.87
M_1	$0+172^{28}$	892.55
9	$0+179^{89}$	892.00
TS	$0+185^{81}$	891.5
H2	$0+200^{00}$	890.85
$Φ_2$	$0+211^{52}$	890.79
10	$0+229^{75}$	892.00
SC	$0+237^{24}$	892.8
11	$0+248^{32}$	894.00
12	$0+262^{17}$	894.00
13	$0+273^{89}$	893.01
B2	$0+285^{61}$	893.32
H3	$0+300^{00}$	893.53
14	$0+318^{39}$	892.00
15	$0+326^{56}$	890.00
CS	$0+333^{99}$	888.69
16	$0+348^{31}$	886.00

T_2	$0+359^{70}$	885.46
17	$0+379^{85}$	885.40
ST	$0+385^{42}$	885.26
M_2	$0+398^{95}$	885.33
K_2	$0+412^{49}$	885.24
18	$0+419^{53}$	885.19
19	$0+435^{46}$	885.57
20	$0+451^{79}$	886.00
21	$0+467^{25}$	885.4
22	$0+482^{71}$	884.00
H5	$0+500^{00}$	882.85
T_1	$0+507^{28}$	882.49
23	$0+532^{36}$	882.00
24	$0+537^{76}$	881.66
25	$0+551^{91}$	882.00
26	$0+562^{79}$	882.00
C	$0+574^{48}$	880.00

4- PRECISE AXIS CALCULATION

	X	Y	ΔX	ΔY	Distance
A	97.87	339.78	75.1	39.22	84.72 m
S_1	172.97	300.56	246.07	66.53	254.91 m
S_2	419.04	367.09			
C	173.30	600.96	245.74	233.87	339.24 m

$$\alpha = \tan^{-1}\frac{75.1}{39.22} = 62.42°$$

$$\beta = \tan^{-1}\frac{246.07}{66.53} = 74.87°$$

$$\gamma = \tan^{-1}\frac{245.74}{233.87} = 46.42°$$

$$\Delta_1 = 180° - \alpha - \beta = 180° - 62.42° - 74.87° = 42.71°$$

$$\Delta_2 = \beta + \gamma = 74.87° + 46.42° = 121.29°$$

$|AS_1| = \sqrt{\Delta X^2 + \Delta Y^2} = \sqrt{75.1^2 + 39.22^2}$ =84.72 m
$|S_1S_2| = \sqrt{\Delta X^2 + \Delta Y^2} = \sqrt{246.07^2 + 66.53^2} = 254.91$ m
$|S_2C| = \sqrt{\Delta X^2 + \Delta Y^2} = \sqrt{245.74^2 + 233.87^2} = 339.24$ m

Horizontal curve 1 calculations using the precise axis method:

$\Delta_1 = 42.71°$

$R_1 = 100\ m$

$t_1 = R_1 \tan(\frac{\Delta_1}{2}) = 100 \tan(\frac{42.71}{2}) = 39.10\ m$

$b_1 = R_1(\sec\left(\frac{\Delta_1}{2}\right) - 1) = 100(\sec\left(\frac{42.71}{2}\right) - 1) = 7.37\ m$

$D_1 = \frac{2\pi \times R_1 \times \Delta_1}{360} = \frac{2\pi \times 100 \times 42.71}{360} = 74.54\ m$

Horizontal curve 2 calculations using the precise axis method:

$\Delta_2 = 121.29°$

$R_2 = 70\ m$ (special case)

$t_2 = R_2 \tan(\frac{\Delta_2}{2}) = 70 \tan(\frac{121.29}{2}) = 124.46\ m$

$b_2 = R_2(\sec\left(\frac{\Delta_2}{2}\right) - 1) = 70(\sec\left(\frac{121.29}{2}\right) - 1) = 72.79\ m$

$D_2 = \frac{2\pi \times R_2 \times \Delta_2}{360} = \frac{2\pi \times 70 \times 121.29}{360} = 148.18\ m$

Km values of special points using the precise axis method:

$A_{km} = 0{+}000^{00}$

$\Phi_{1,km} = A_{km} + |AS_1| - t_1 = 0.00 + 84.72 - 39.10 = 0{+}045^{62}$

$B_{1,km} = \Phi_{1,km} + \frac{D_1}{2} = 45.62 + \frac{74.54}{2} = 0{+}082^{89}$

$\Gamma_{1,km} = \Phi_{1,km} + D_1 = 45.62 + 74.54 = 0{+}120^{16}$

$\Phi_{2,km} = \Gamma_{1,km} + |S_1S_2| - t_1 - t_2 = 120.16 + 254.91 - 39.10 - 124.46 = 0{+}211^{51}$

$B_{2,km} = \Phi_{2,km} + \frac{D_2}{2} = 211.51 + \frac{148.18}{2} = 0{+}285^{60}$

$\Gamma_{2,km} = \Phi_{2,km} + D_2 = 221.51 + 148.18 = 0{+}359^{69}$

$C_{km} = |S_2C| + \Gamma_{2,km} - t_2 = 339.24 + 359.69 - 124.46 = 0{+}574^{47}$

5- TRANSITION CURVE CALCULATIONS

Although the start of my first horizontal curve is more than 40 meters away from the A point (as required in the 1st part of the highway project "Do not place horizontal curve closer than 40 m to the beginning and end points of the highway"), I have calculated that when applying the ramp, K_1 is 5.65 meters behind the A point becuase Lg for the 1st curve was calculated as 49 meters and K = 26.78 meters, TS was 21.13 meters away from the A point which would make K_1 5.65 meters behind the A point, for this reason, I have chosen to place the transition curve for the second curve.

The transition curve calculations are made for the second curve (R = 70 m)

Superelevation:

Maximum allowable superelevation is given as 8%

$$q_{pra} = 0.443 \times \frac{V_p^2}{R} = 0.443 \times \frac{70^2}{70} = 31.01 > 8 \qquad improper!$$

Therefore select the superelevation as 8% and limit the speed.

New project speed:

$$8 = 0.443 \times \frac{V_p^2}{70} \quad \rightarrow \quad V_p = 35.55\ m/s$$

Selection of the clothoid parameter:

Optical condition:

$$\frac{R}{3} \leq A \leq R \ \rightarrow \ \frac{70}{3} \leq A \leq 70$$

Dynamic condition:

$$A_{min} = 0.19 \times \sqrt{V_p^3} = 0.19 \times \sqrt{35.55^3} = 40.27\ m$$

No need for superelevation condition.

Minimum length control:

L_g = At least 45 m

$$L_g = \frac{A^2}{R} \rightarrow L_{g,min} = \frac{A_{min}^2}{R} = 45 \rightarrow 45 = \frac{A_{min}^2}{70} \rightarrow A_{min} = 56.12\ m$$

A must be a multiple of 5, we chose the value of **A = 60 m** in order to be safer

$$L_g = \frac{A^2}{R} = \frac{60^2}{70} = 51.43\ m$$

Geometrical control:

$$\frac{L_g}{2} \leq \frac{D}{2} \rightarrow \frac{51.43}{2} \leq \frac{148.18}{2}$$

Jerk control:

$$p' = \frac{V_p^3}{46.7 \times R \times L_g} = \frac{35.55^3}{46.7 \times 70 \times 51.43} = 0.267\ m/s^3 < 0.6\ m/s^3$$

L (m)	X (m)	Y (m)
5	5.0000	0.0058
10	9.9998	0.0463
15	14.9985	0.1562
20	19.9938	0.3703
25	24.9812	0.7230
30	29.9532	1.2486
35	34.8988	1.9808
40	39.8029	2.9525
45	44.6453	4.1949
50	49.4005	5.7370
51.43	50.7402	6.2369

Klotoid elemanlarının hesabında ise aşağıdaki formüller kullanılır.

$$X = L - \frac{L^5}{40 \cdot A^4} + \frac{L^9}{3456 \cdot A^8} - \cdots \quad (m)$$

$$Y = \frac{L^3}{6 \cdot A^2} - \frac{L^7}{336 \cdot A^6} + \frac{L^{11}}{42240 \cdot A^{10}} - \cdots \quad (m)$$

$$\tau = \frac{L}{2 \cdot R} \quad \text{(radyan)(dereceye de çevrilecek)}$$

$$X_m = X - R \cdot \sin\tau \quad (m)$$

$$Y_m = Y + R \cdot \cos\tau \quad (m)$$

$$T_u = X - Y \cdot \mathrm{cotg}\tau \quad (m)$$

$$T_k = \frac{Y}{\sin\tau} \quad (m)$$

$$\Delta R = Y_m - R \quad (m)$$

$$S = \sqrt{X^2 + Y^2} \quad (m)$$

$$\sigma = \mathrm{arctg}(\frac{Y}{X}) \quad (m)$$

$$\overline{TS} = X_m + (R + \Delta R) \cdot \mathrm{tg}\frac{\Delta}{2} \quad (m)$$

$$\Delta C = \Delta - 2 \cdot \tau \quad (\text{derece})$$

$$Dc = \frac{2 \cdot \pi \cdot R \cdot \Delta C}{360} \quad (m)$$

- R = 70 m
- L_g = 51.43 m
- A = 60 m
- X = 50.7402 m
- Y = 6.2369 m
- τ = 0.367 rad = 21.038˚
- X_m = 25.6111 m
- Y_m = 71.5709 m
- T_u = 34.5247 m
- T_k = 17.3736 m
- ΔR = 1.5709 m
- S = 51.1221m
- σ = 7.008˚
- $\overline{TS}$ = 152.8626 m
- ΔC = 79.214˚
- Dc = 96.7782 m

<u>Starting and end points of the clothoid:</u>

$TS_{km} = \Phi_{2,km} - L_g/2 = 211.52 - 25.715 = 0+185^{81}$
$SC_{km} = \Phi_{2,km} + L_g/2 = 211.52 + 25.715 = 0+237^{24}$
$CS_{km} = \Gamma_{2,km} - L_g/2 = 359.70 - 25.715 = 0+333^{99}$
$ST_{km} = \Gamma_{2,km} + L_g/2 = 259.70 + 25.715 = 0+385^{42}$

6- WIDENING AND RAMP LENGTH CALCULATIONS

Widening:

R<200 m so there is need for expansion. Expansion is equal on both sides of the road and it will expand linearly throughout the clothoid.

$$b_g = \frac{n \times l^2}{2 \times R} + \frac{0.05 \times V_p}{\sqrt{R}} = \frac{2 \times 12^2}{2 \times 70} + \frac{0.05 \times 35.55}{\sqrt{70}} = 2.270\ m > 2\ m \quad improper!$$

As a result, we need to take the value of b_g as the maximum value for the expansion:

b_g = 2 m

Both sides of the road will expand equally and therefore it will start from zero and increase linearly until it reaches maximum expansion at SC point

$B_g(TS) = 0$

$B_g(\Phi_1) = b_g/4 = 2/4 = 0.5$ m

$B_g(SC) = b_g/2 = 2/2 = 1$ m

Ramp length:

h_1 is the elevation difference at TS (no expansion here yet only roof slope). h_2 is the elevation difference at SC point where the superelevation and expansion reach their maximum.

$$h_1 = q_1 \times \frac{b}{2} = 0.02 \times \frac{10}{2} = 0.1\ m$$

$$h_2 = q_2 \times \frac{b + b_g}{2} = 0.08 \times \frac{10 + 2}{2} = 0.48\ m$$

$$K = \frac{2 \times L_g \times h_1}{h_2 - h_1} = \frac{2 \times 51.43 \times 0.1}{0.48 - 0.1} = 27.068\ m$$

Km values of critical points:

TS_{km} = 0+185[81]
SC_{km} = 0+237[24]
CS_{km} = 0+333[99]
ST_{km} = 0+385[42]

$K_{1,km} = TS_{km} - K$ = 185.81 – 27.068 = 0+158[74]

$K_{2,km} = ST_{km} + K = 385.42 + 27.068 = 0+412^{49}$

$M_{1,km} = TS_{km} – K/2 = 185.81 – 13.534 = 0+172^{28}$

$M_{2,km} = ST_{km} + K/2 = 385.42 + 13.534 = 0+398^{95}$

$b_g(TS) = 0$

$b_g(H_2) = (H_{2,km} – TS_{km})*(2/51.43) = 0.552$ m

$b_g(\Phi_2) = (\Phi_{2,km} – TS_{km})*(2/51.43) = 1.000$ m

$b_g(10) = (10_{km} – TS_{km})*(2/51.43) = 1.709$ m

$b_g(SC) = 2$ m

$h_8 = ((0.1+0.48)/78.498)*(7.75) – 0.1 = -0.043$ m

$h_9 = ((0.1+0.48)/78.498)*(21.15) – 0.1 = 0.056$ m

$h_{TS} = ((0.1+0.48)/78.498)*(27.07) – 0.1 = 0.100$ m

$h_{H2} = ((0.1+0.48)/78.498)*(41.26) – 0.1 = 0.205$ m

$h_{\Phi 2} = ((0.1+0.48)/78.498)*(52.78) – 0.1 = 0.290$ m

$h_{10} = ((0.1+0.48)/78.498)*(71.01) – 0.1 = 0.425$ m

$q(8) = \left(\frac{4}{27.08} \times 7.75\right) - 2 = -0.86\%$

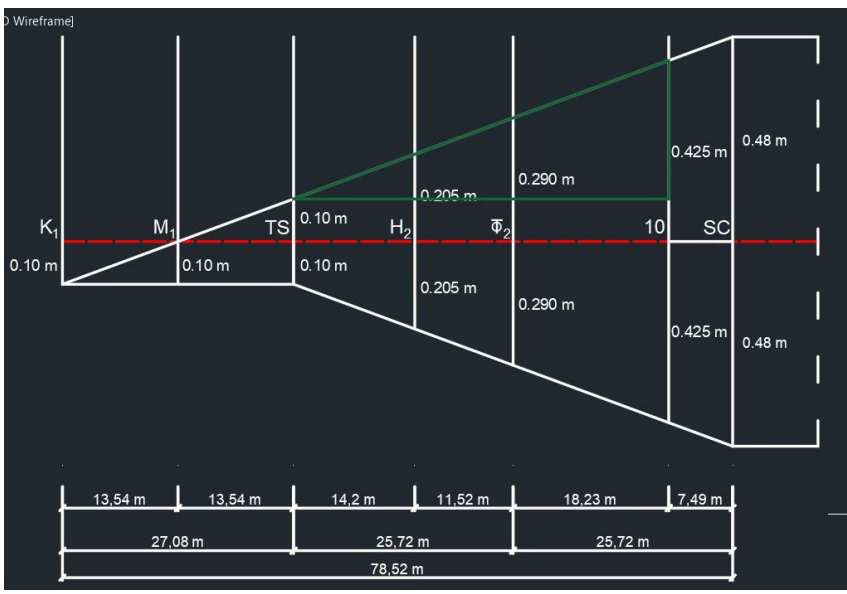
$q(9) = \left(\frac{4}{27.08} \times 21.15\right) - 2 = 1.12\%$

I calculated the supperelevation by drawing a triangle from point TS to the desired point (8%-2% = 6%), see the green triangle.

$q(H_2) = \left(\frac{8-2}{51.44} \times 14.2\right) + 2 = 3.66\%$

$q(\Phi_2) = \left(\frac{8-2}{51.44} \times 25.72\right) + 2 = 5\%$

$q(10) = \left(\frac{8-2}{51.44} \times 43.95\right) + 2 = 7.13\%$



Cross section	Km	Distance form the beginning (m)	Superelevation %		Elevation difference with respect to the axis elevation		
			Outer Edge	Inner Edge	Outer Edge	Centerline	Inner Edge
K_1	$0+158^{74}$	**0**	**-2**	**-2**	**-0.100**	**0**	**-0.100**
8	$0+166^{49}$	**7.75**	**-0.86**	**-2**	**-0.043**	**0**	**-0.100**
M_1	$0+172^{28}$	**13.54**	**0**	**-2**	**0.000**	**0**	**-0.100**
9	$0+179^{89}$	**21.15**	**1.12**	**-2**	**0.056**	**0**	**-0.100**
TS	$0+185^{81}$	**27.07**	**2**	**-2**	**0.100**	**0**	**-0.100**
H_2	$0+200^{00}$	**41.26**	**3.66**	**-3.66**	**0.205**	**0**	**-0.205**
Φ_2	$0+211^{52}$	**52.78**	**5**	**-5**	**0.290**	**0**	**-0.290**
10	$0+229^{75}$	**71.01**	**7.13**	**-7.13**	**0.425**	**0**	**-0.425**
SC	$0+237^{24}$	**78.5**	**8**	**-8**	**0.480**	**0**	**-0.480**

7- VERTICAL CURVE CALCULATIONS

$S_{AC}\% = 3.48\%$

$g_1 = -3.48\%$

$g_2 = 1\%$ (last digit of the student number is zero, n=1)

$G = |-3.48-1| = 4.48\% = 0.0448 > 0.5\%$ proper!

$R = R_{sag} = 3000$ m

$$K = \frac{1}{2 \times R} = \frac{1}{2 \times 3000} = 0.00017\ m$$

$$t = \frac{R \times G}{2} = \frac{3000 \times 0.0448}{2} = 67.2\ m$$

$L = 2t = 134.4$ m > 120 m proper!

$$e = \frac{G \times L}{8} = \frac{0.0448 \times 134.4}{8} = 0.753\ m$$

$P_{km} = 0+574^{48}$
$P_{kot} = 880$ m

$T_{1,km} = P_{km} - t = 574.48 - 67.2 = 0+507^{28}$
$T_{2,km} = P_{km} + t = 574.48 + 67.2 = 0+641^{68}$

$B_{km} = P_{km} = 0+574^{48}$

$$DN_{km} = T_{1,km} + \frac{g_1 \times L}{G} = 507.28 + \frac{0.0348 \times 134.4}{0.0448} = 0 + 611^{68}$$

$T_{1,kot} = P_{kot} – g_1*t = 880 – (–2.339) = 882.339$ m
$T_{2,kot} = P_{kot} – g_2*t = 880 – (0.672) = 879.328$ m
$B_{kot} = P_{kot} + e = 880 + 0.753 = 880.753$ m

$$DN_{kot} = T_{1,kot} + \frac{g_1^2 \times L}{2G} = 882.339 - 1.81656 = 880.52244\ m$$

Cross section	Km	x (m)	r =g_1.x	T_{1kot}-r	y=k.x^2	T_{1kot}-r+y
T_1	**507.28**	**0**	**0**	**882.339**	**0.00000**	**882.3390**
23	**523.36**	**16.08**	**0.55958**	**881.779**	**0.04396**	**881.8234**
24	**537.76**	**30.48**	**1.06070**	**881.278**	**0.15794**	**881.4362**
25	**551.91**	**44.63**	**1.55312**	**880.786**	**0.33861**	**881.1245**
26	**562.79**	**55.51**	**1.93175**	**880.407**	**0.52383**	**880.9311**
C	**574.48**	**67.2**	**2.33856**	**880.000**	**0.76769**	**880.7681**
DN	**611.68**	**104.4**	**3.63312**	**878.706**	**1.85289**	**880.5588**
T_2	**641.68**	**134.4**	**4.67712**	**877.662**	**3.07077**	**880.7327**

8- CALCULATION OF RED ELEVATION ON PROFILE

Cross section no	Km	Black Elevation	Red Elevation	Cut Height	Fill Height
A	$0+000^{00}$	900.00	900	0	0
1	$0+025^{00}$	901.87	899.13	2.74	0
Φ_1	$0+045^{63}$	900.63	898.41	2.22	0
2	$0+052^{27}$	900.00	898.18	1.82	0
3	$0+060^{60}$	898.00	897.89	0.11	0
4	$0+071^{75}$	896.58	897.50	0	0.92
B1	$0+082^{90}$	896.05	897.11	0	1.06
H1	$0+100^{00}$	893.87	896.52	0	2.65
5	$0+110^{08}$	893.05	896.17	0	3.12

Ŧ$_1$	0+120^{16}	892.20	895.82	0	3.62
6	0+137^{92}	890.83	895.20	0	4.37
7	0+153^{09}	892.00	894.67	0	2.67
K$_1$	0+158^{74}	892.83	894.47	0	1.64
8	0+166^{49}	892.87	894.20	0	1.33
M$_1$	0+172^{28}	892.55	894.00	0	1.45
9	0+179^{89}	892.00	893.74	0	1.74
TS	0+185^{81}	891.5	893.53	0	2.03
H2	0+200^{00}	890.85	893.04	0	2.19
ⱷ$_2$	0+211^{52}	890.79	892.64	0	1.85
10	0+229^{75}	892.00	892.00	0	0
SC	0+237^{24}	892.8	891.74	1.06	0
11	0+248^{32}	894.00	891.35	2.65	0
12	0+262^{17}	894.00	890.87	3.13	0
13	0+273^{89}	893.01	890.46	2.55	0
B2	0+285^{61}	893.32	890.06	3.26	0
H3	0+300^{00}	893.53	889.56	3.97	0
14	0+318^{39}	892.00	888.92	3.08	0
15	0+326^{56}	890.00	888.63	1.37	0
CS	0+333^{99}	888.69	888.37	0.32	0
16	0+348^{31}	886.00	887.87	0	1.87
Ŧ$_2$	0+359^{70}	885.46	887.48	0	2.02
17	0+379^{85}	885.40	886.78	0	1.38
ST	0+385^{42}	885.26	886.58	0	1.32
M$_2$	0+398^{95}	885.33	886.11	0	0.78
K$_2$	0+412^{49}	885.24	885.64	0	0.40
18	0+419^{53}	885.19	885.39	0	0.20
19	0+435^{46}	885.57	884.84	0.73	0
20	0+451^{79}	886.00	884.27	1.73	0
21	0+467^{25}	885.4	883.73	1.67	0
22	0+482^{71}	884.00	883.19	0.81	0
H5	0+500^{00}	882.85	882.59	0.26	0
T$_1$	0+507^{28}	882.49	882.34	0.15	0
23	0+532^{36}	882.00	881.78	0.18	0

24	$0+537^{76}$	881.66	881.28	0.22	0
25	$0+551^{91}$	882.00	880.79	0.88	0
26	$0+562^{79}$	882.00	880.41	1.07	0
C	$0+574^{48}$	880.00	880.00	0	0.77

9- GEOMETRIC VALUES FOR DRAWING WALL

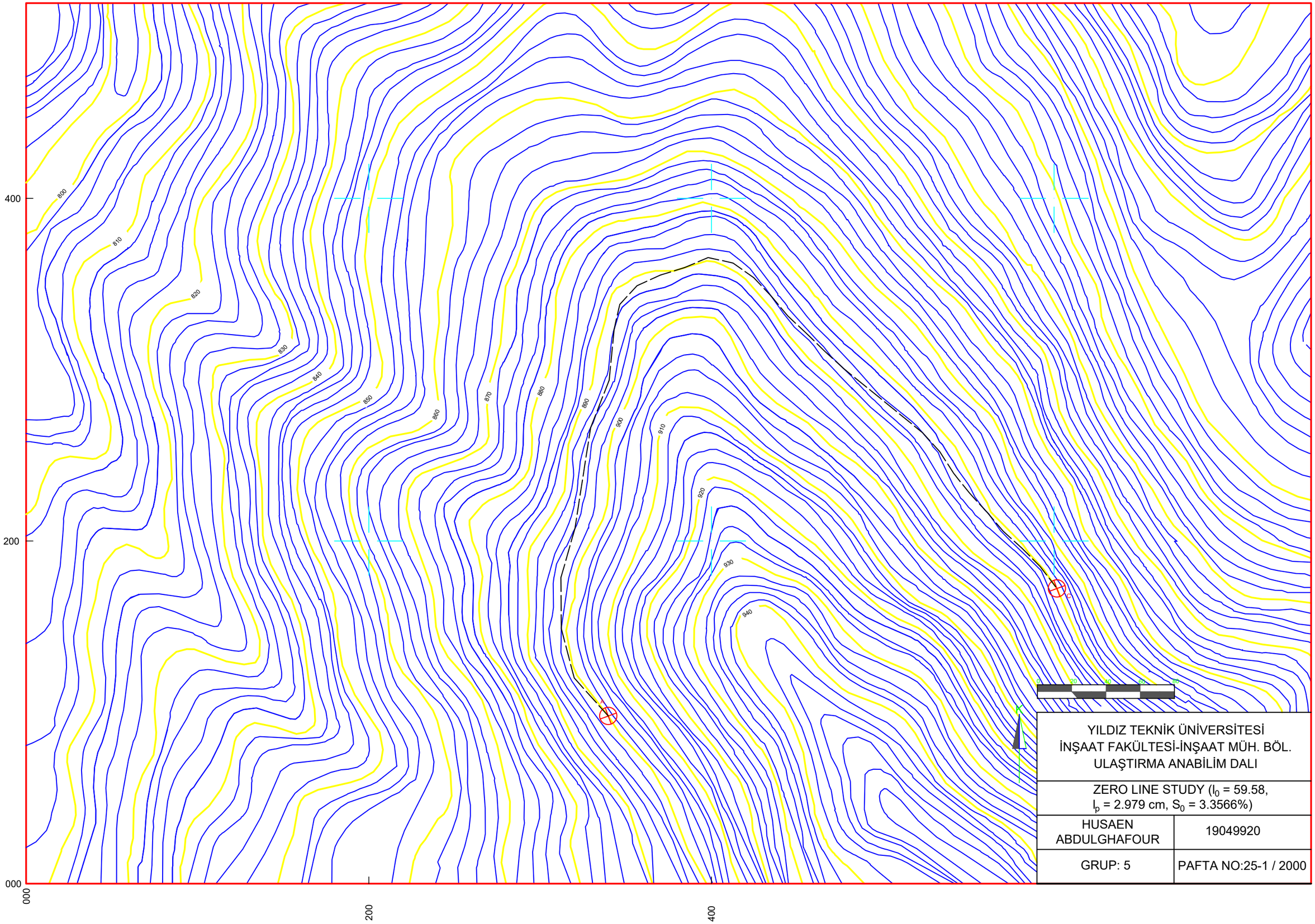
I honestly do not know how to calculate them.

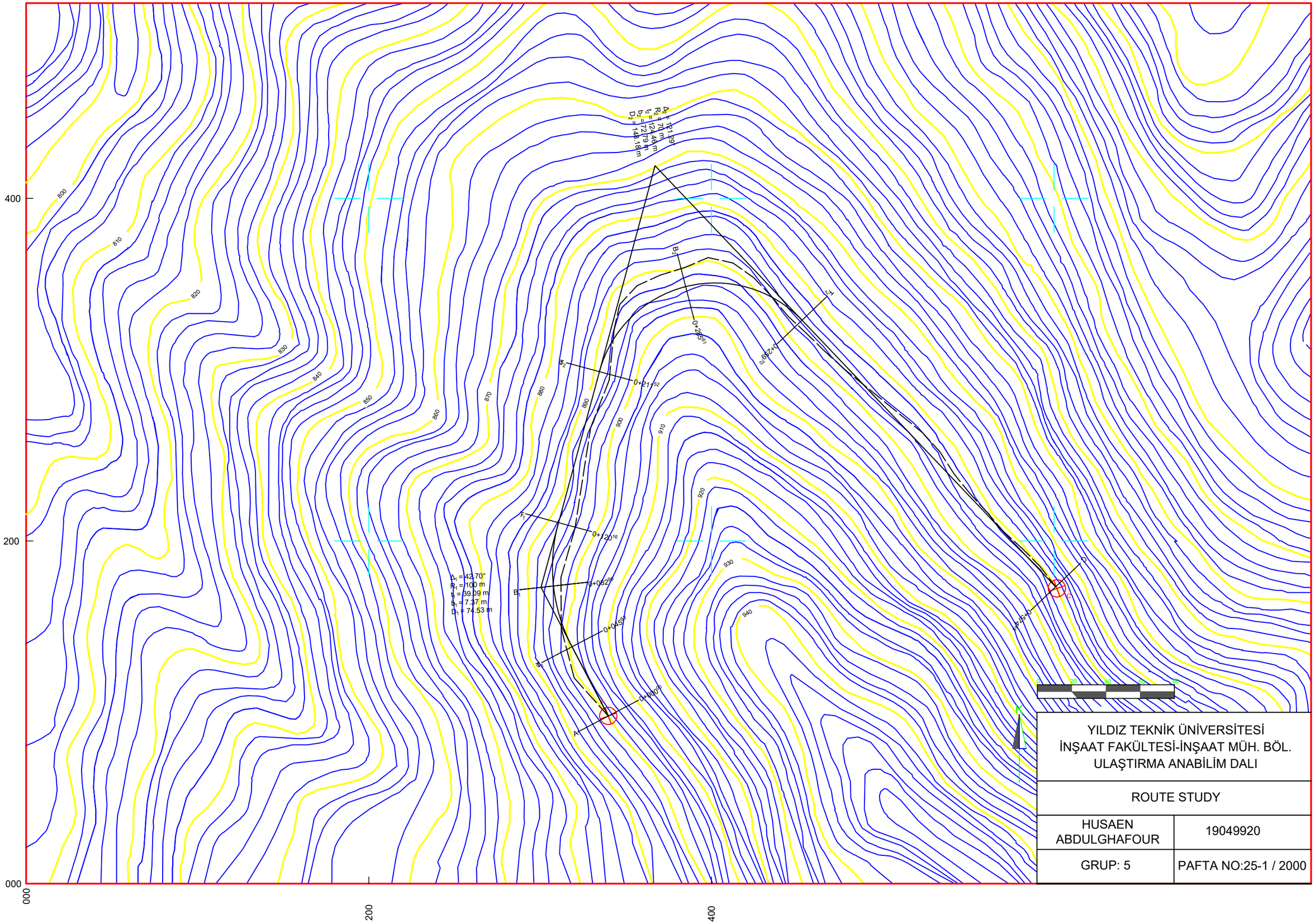
10- AREA CALCULATIONS OF CROSS-SECTIONS

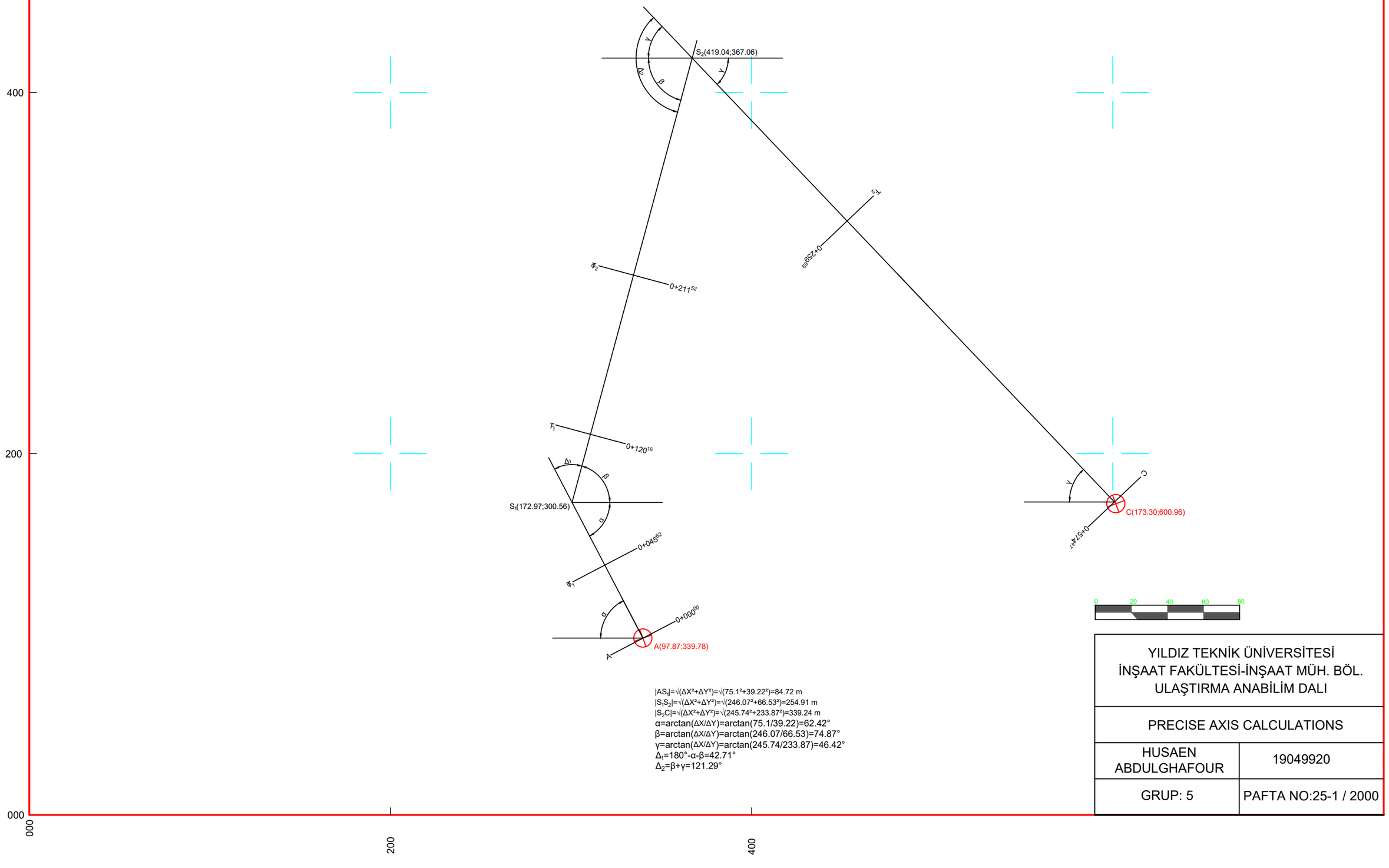
I just drew the typical cross section but unfortunately, I couldn't draw the other cross sections.

11- CONCLUSION

The highway connecting points A and C was made according to the rules and specifications outlined in the class. The highway was designed to meet the criteria of comfort for drivers and be economically feasible at the same time. The cut and fill earthworks were almost balanced wich would reduce the cost of construction.
Overall, the project helped me get an insight into how highways are designed and constructed and it made me appreciate the work and the effort of highway engineers in designing safe, comfortable, and economical highways.


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80
K
YILDIZ TEKNİK ÜNİVERSİTESİ
İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL.
ULAŞTIRMA ANABİLİM DALI
ZERO LINE STUDY (l_0 = 59.58, l_p = 2.979 cm, S_0 = 3.3566%)
HUSAEN ABDULGHAFOUR
19049920
GRUP: 5
PAFTA NO:25-1 / 2000


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930
940
Δ_2 = 121.29°
R_2 = 70 m
t_2 = 124.46 m
b_2 = 72.79 m
D_2 = 148.18 m
B_2
T_2
0+229.70
0+229.51
A_2
0+121.82
T_1
0+120.16
Δ_1 = 42.70°
R_1 = 100 m
t_1 = 39.09 m
b_1 = 7.37 m
D_1 = 74.53 m
B_1
0+082.98
0+045.98
A_1
0+000.00
D
C
0+425.10
0
20
40
60
80
K
YILDIZ TEKNİK ÜNİVERSİTESİ
İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL.
ULAŞTIRMA ANABİLİM DALI
ROUTE STUDY
HUSAEN ABDULGHAFOUR
19049920
GRUP: 5
PAFTA NO:25-1 / 2000



$|AS_1|=\sqrt{(\Delta X^2+\Delta Y^2)}=\sqrt{(75.1^2+39.22^2)}=84.72$ m

$|S_1S_2|=\sqrt{(\Delta X^2+\Delta Y^2)}=\sqrt{(246.07^2+66.53^2)}=254.91$ m

$|S_2C|=\sqrt{(\Delta X^2+\Delta Y^2)}=\sqrt{(245.74^2+233.87^2)}=339.24$ m

$\alpha=\arctan(\Delta X/\Delta Y)=\arctan(75.1/39.22)=62.42°$

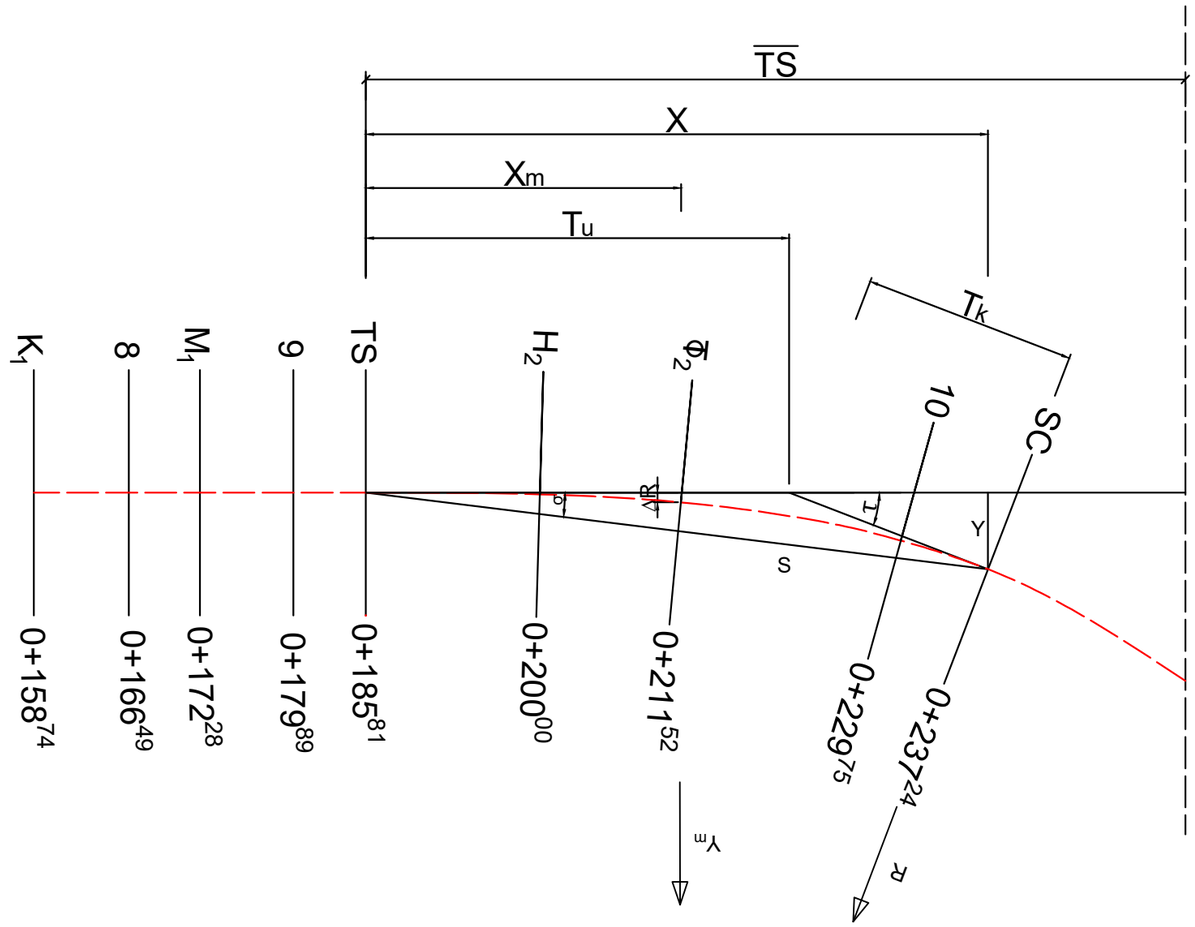
$\beta=\arctan(\Delta X/\Delta Y)=\arctan(246.07/66.53)=74.87°$

$\gamma=\arctan(\Delta X/\Delta Y)=\arctan(245.74/233.87)=46.42°$

$\Delta_1=180°-\alpha-\beta=42.71°$

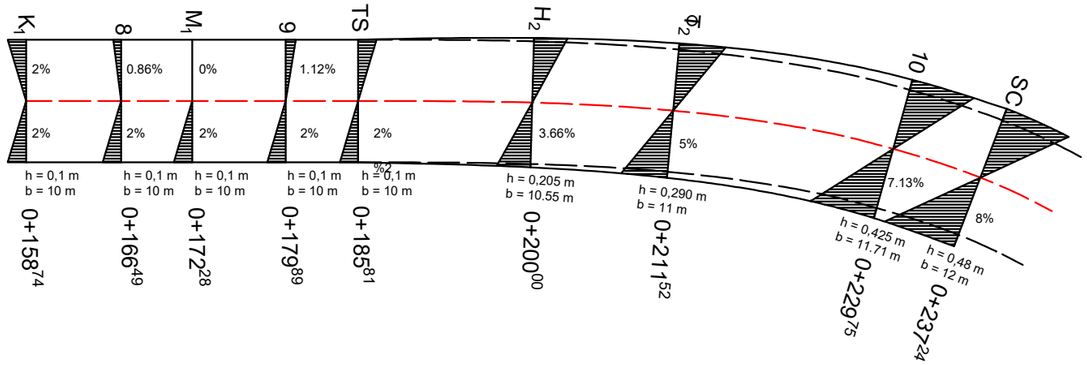
$\Delta_2=\beta+\gamma=121.29°$

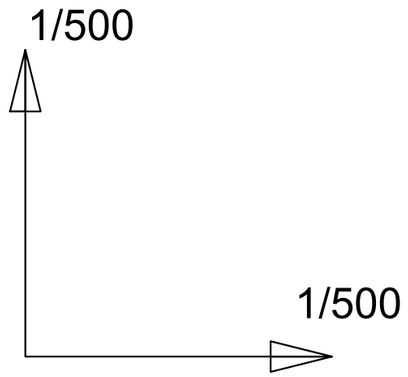
YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
PRECISE AXIS CALCULATIONS	
HUSAEN ABDULGHAFOUR	19049920
GRUP: 5	PAFTA NO:25-1 / 2000

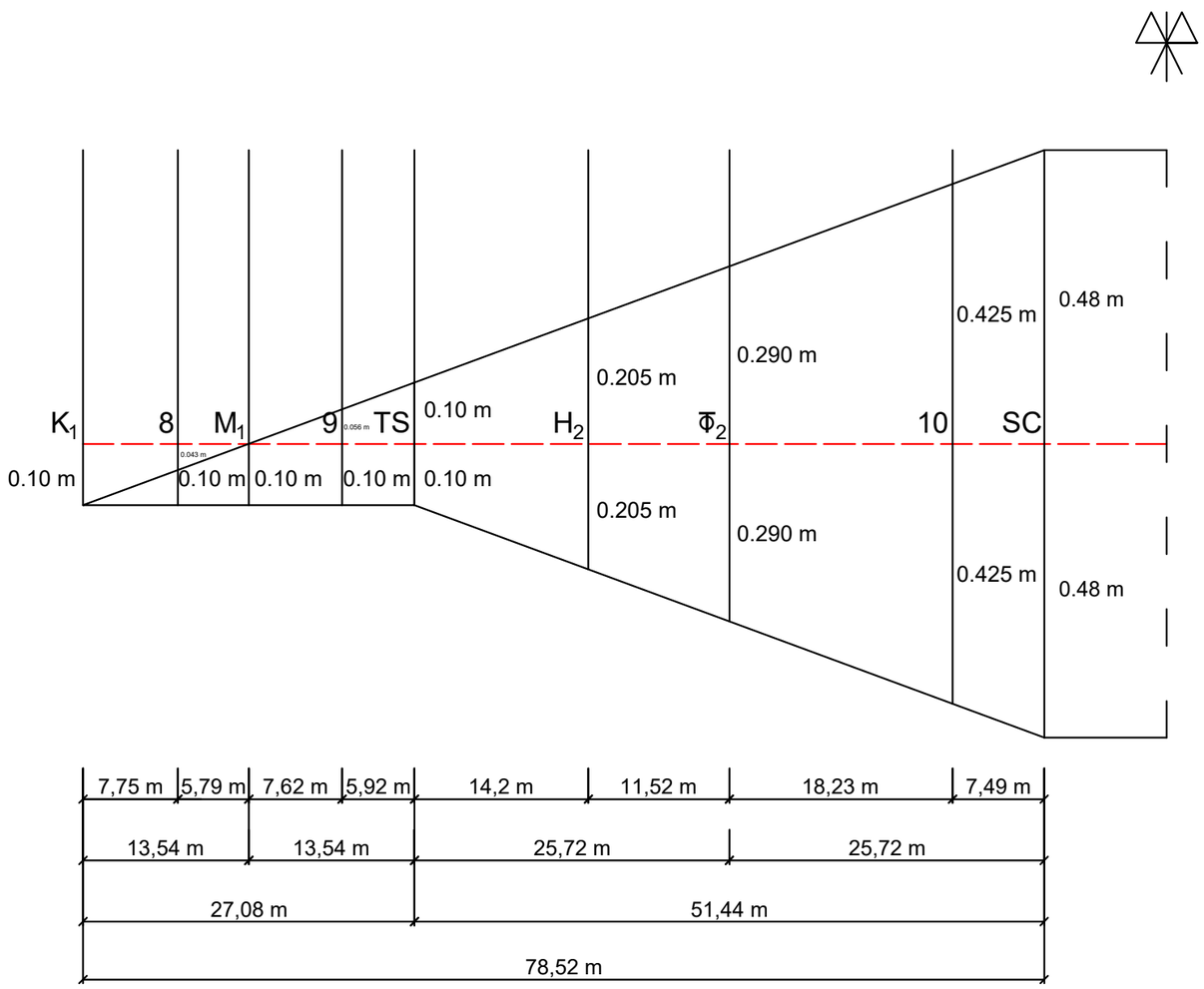


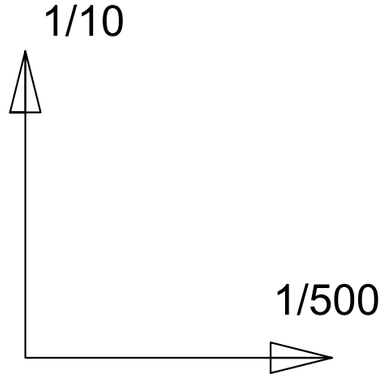
R =70 m
L_g= 51.43 m
A = 60 m
X = 50.7402 m
Y = 6.2369 m
τ = 0.367 rad = 21.038°
X_m = 25.6111 m
Y_m = 71.5709 m
T_u = 34.5247 m
T_k= 17.3736 m
ΔR = 1.5709 m
S = 51.1221 m
σ = 7.008°m
$\overline{TS}$= 152.8626 m
ΔC = 79.214°
DC = 96.7782 m

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CLOTHOID DETAILS	
HUSAEN ABDULGHAFOUR	19049920
GRUP: 5	SCALE:1/500

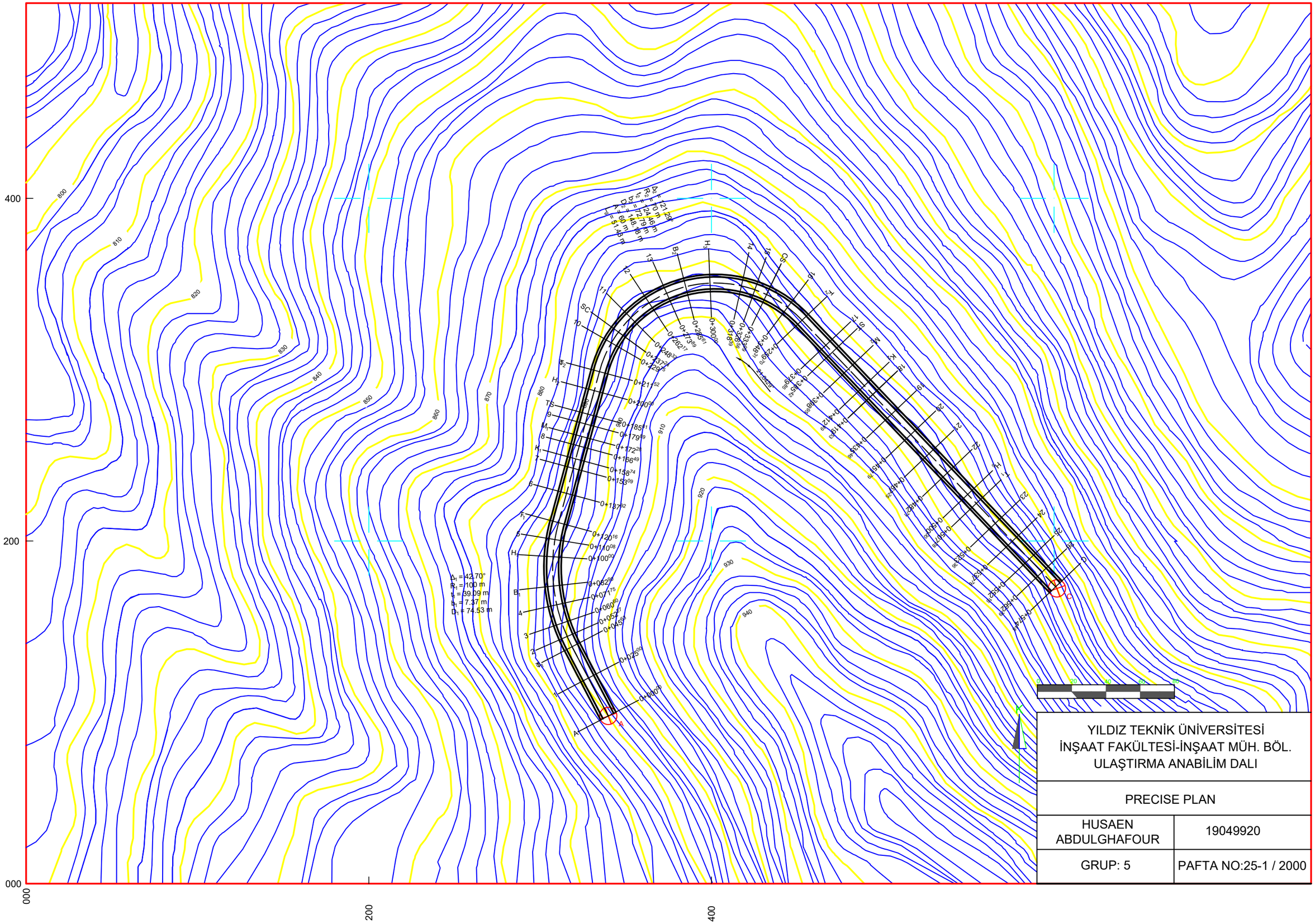




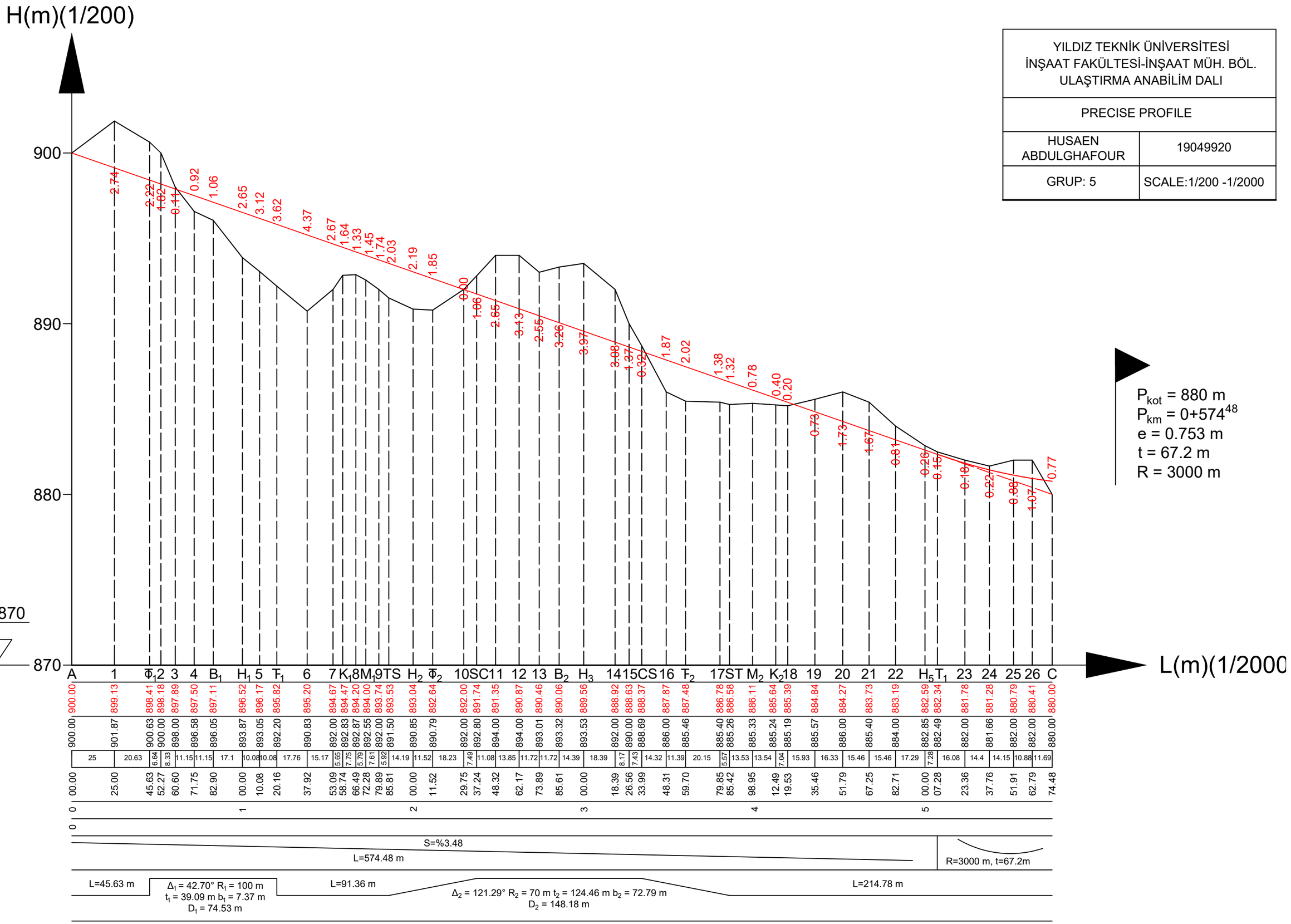




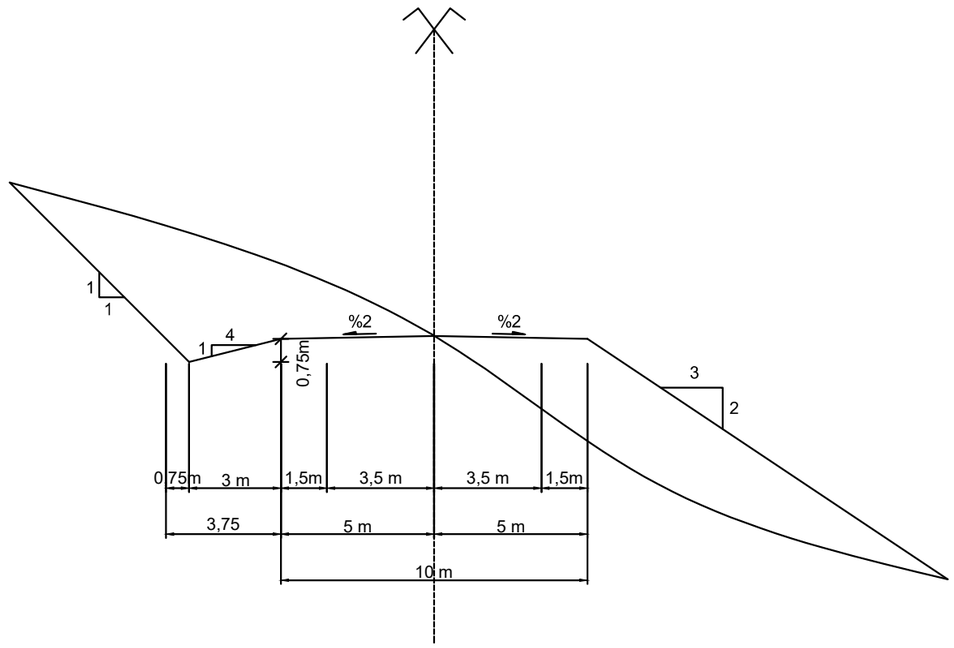
YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
SUPERELEVATION DETAILS	
HUSAEN ABDULGHAFOUR	19049920
GRUP: 5	SCALE:1/10 - 1/500


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850
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870
880
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920
930
940
Δ1 = 42.70°
R1 = 100 m
t1 = 39.09 m
h1 = 7.37 m
D1 = 74.53 m
Δ2 = 70.74°
R2 = 24.46 m
t2 = 17.79 m
h2 = 148.8 m
D2 = 60 m
A = 51.48 m
0 20 40 60 80
N
YILDIZ TEKNİK ÜNİVERSİTESİ
İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL.
ULAŞTIRMA ANABİLİM DALI
PRECISE PLAN
HUSAEN ABDULGHAFOUR
19049920
GRUP: 5
PAFTA NO:25-1 / 2000

YILDIZ TEKNİK ÜNİVERSİTESİ İNŞAAT FAKÜLTESİ-İNŞAAT MÜH. BÖL. ULAŞTIRMA ANABİLİM DALI	
PRECISE PROFILE	
HUSAEN ABDULGHAFOUR	19049920
GRUP: 5	SCALE:1/200 -1/2000



THE TYPICAL CROSS SECTION



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THE TYPICAL CROSS SECTION	
HUSAEN ABDULGHAFOUR	19049920
GRUP: 5	SCALE:1/200